

Python Resources

Welcome to 'Python Programming Masterclass' Resource Section.

Quick Note: This resource section only makes sense if you have watched the course lectures otherwise you will see random points and multiple links.

Includes: Reference links, reading material, cheatsheet PDFs, exercises and more.

Official Websites

- Python: <https://www.python.org/>
- Python Package Index: <https://pypi.org/>

AI Assistance

- ChatGPT: <https://chat.openai.com/>
- Gemini: <https://gemini.google.com/>
- Claude: <https://claude.ai/>

Code Editor

- VS Code: <https://code.visualstudio.com/>
 - Extensions:
 - Python Extension Pack, Material Icon Theme
 - Theme (File > Preferences > Color Theme)
 - Icons (File > Preferences > File Icon Theme)
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Notes

- Python Concepts Notes: <https://unwiredlearning.com/blog/python>
 - Flask Notes: <https://unwiredlearning.com/blog/flask>
 - Pandas Practice Files / Data Analysis:
<https://github.com/ShubhamSarda/pandas-resources/blob/main/README.md>
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Projects

Face Detection / OpenCV:

- Source Code: <https://github.com/ShubhamSarda/OpenCV-Basics>
- Library: <https://pypi.org/project/opencv-python/>
- Haarcascade File:
<https://drive.google.com/drive/folders/1w-kORB1g08mkt0CJnIIN7ePggGMVI7Y7>

Password Generator / random:

- Library: <https://docs.python.org/3/library/random.html>

Data Analysis / Pandas:

- Official Website: <https://pandas.pydata.org/>
- Pandas Basics Cheatsheet: https://pandas.pydata.org/Pandas_Cheat_Sheet.pdf

GUI Portfolio Application / Tkinter:

- Source Code: <https://github.com/ShubhamSarda/Python-GUI-Programming-With-Tkinter>
- Example API Link Structure / *(Note: Update Your Own API Here):*
https://pro-api.coinmarketcap.com/v1/cryptocurrency/listings/latest?start=1&limit=2&convert=USD&CMC_PRO_API_KEY=8decb82f-10f0-4c6c-8b0a-79d97b8cc67a
- JSON Formatter: <https://codebeautify.org/jsonviewer>
- SQLite Datatypes: <https://www.sqlite.org/datatype3.html>
- DB Browser for SQLite: <https://sqlitebrowser.org/>

Web Scraping Bot / BeautifulSoup:

- Source Code: <https://github.com/ShubhamSarda/WebScrapeBot>

Automated Weather Forecast / Pillow:

- Source Code: <https://github.com/ShubhamSarda/WeatherForecastPost>
- Practice Image:
<https://raw.githubusercontent.com/ShubhamSarda/WeatherForecastPost/main/post.png>
- Exercise Image:
<https://raw.githubusercontent.com/ShubhamSarda/WeatherForecastPost/main/story.png>

Twitter Bot / Tweepy 3.7:

- Source Code: <https://github.com/ShubhamSarda/twbot>
- Twitter Developer Account: <https://developer.twitter.com/en>
- Tweepy Documentation: <https://docs.tweepy.org/en/stable/index.html>
- **Q:** What to answer when asked about why we need an API key on Twitter Developer Dashboard?
A: Just let them know that you require the API key for educational purpose as a student and don't have any commercial use, don't copy-paste any answer, don't attach any link or image, and don't make any reference name in the answer. Make sure to have an original reply otherwise, the bot will detect common internet answers.

Here is a common answer,

I will use Twitter API for my academic project where I am analyzing user tweets and replies. I am a computer science university student and don't have any commercial use. Just using it for educational purposes.

Podcast Script Writer / OpenAI:

- Source Code: <https://github.com/ShubhamSarda/minigpt>
- Official Website: <https://openai.com/>
- Playground: <https://platform.openai.com/playground>
- API Key: <https://platform.openai.com/account/api-keys>
- Documentation: <https://platform.openai.com/docs/guides/completion/introduction>

Conversational Chatbot / OpenAI:

- Source Code: <https://github.com/ShubhamSarda/chatbot>

Image Generator With DALL-E / OpenAI:

- Source Code: <https://github.com/ShubhamSarda/imagegpt>
- DALL-E 2: <https://openai.com/product/dall-e-2>
- Documentation: <https://platform.openai.com/docs/guides/images/introduction>

Video Summarizer / OpenAI / Moviepy :

- Source Code: <https://github.com/ShubhamSarda/summarize>

Portfolio Web Application / Flask:

- Source Code: <https://github.com/ShubhamSarda/portfolyx>
 - Flask Framework Documentation: <https://flask.palletsprojects.com/en/latest/>
 - Images: <https://unsplash.com/>
 - Icons: <https://icons8.com/icons/set/programming>
 - Fonts: <https://fonts.google.com/>
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Assignment Solutions

- Python Image Optimization:
<https://replit.com/@ShubhamSarda/TinyPNGSample#main.py>
 - Web Scraping Encyclopedia Articles:
<https://replit.com/@ShubhamSarda/s-WebScrapingWikipedia#main.py>
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Cheatsheet

- Python Basics:
https://perso.limsi.fr/poinal/_media/python:cours:mementopython3-english.pdf
 - Pandas Basics:
https://pandas.pydata.org/Pandas_Cheat_Sheet.pdf
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Interview Questions

- Common Question:
- <https://www.softwaretestinghelp.com/python/python-interview-questions/>
- Practice with ChatGPT, Gemini or Claude,
Ask for common interview questions for particular topics
prompt: most common interview questions about 'topic' in python along with explanation
- Read Notes: <https://unwiredlearning.com/blog/python>
Since all topics are in sequence, it will help you to revise and prepare for common questions.